

Toward a Duality Theory for Feature and Sample Screening

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Abstract

Feature screening and sample screening reduce the cost of convex optimization by removing, respectively, features or samples before the problem is solved. Although both rely on convex duality, they have been developed largely independently, model by model, and the mathematical relationship between the two paradigms has remained informal. In this paper, we introduce the class of Fenchel–Rockafellar (FR) representations, on which both screening and FR duality are defined as transformations. We then prove that screening is equivariant under FR duality. Thus, the duality between feature screening and sample screening is not merely an analogy between two types of screening rules, but a commutative relation between transformations on FR representations. The proof uses the structural fact that every screening operator factors into a translation followed by a restriction, whose primitive forms are equivariant under FR duality. These results provide a mathematical formulation of the duality between feature and sample screening, laying a foundation for a systematic duality theory of screening in machine learning.

Keywords: convex optimization, Fenchel–Rockafellar duality, sparse learning, Lasso, SVMs

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1 Introduction

[LE: Will Be Verified.] Large-scale convex optimization is central to modern machine learning. In recent years, *screening* methods have emerged as an effective technique for reducing problem size before or during optimization, thereby accelerating the solution of large-scale learning problems [El Ghaoui et al., 2010a].

[LE: Will Be Verified.] Screening methods have been studied from several complementary perspectives. According to whether they guarantee preservation of the optimal solution, they are broadly classified into safe screening methods [El Ghaoui et al., 2010a] and unsafe screening methods [Tibshirani et al., 2012; Fan and Lv, 2018]. According to how screening is performed, they may be static [El Ghaoui et al., 2010a], dynamic [Bonnefoy et al., 2014], or sequential [Wang et al., 2013]. Recent advances mainly focus on constructing tighter safe regions, including ball [Ndiaye et al., 2017; Tran et al., 2025], dome [Tran et al., 2022], and ellipsoid [Dai and Pelekmans, 2012] regions. The screening paradigm has also been generalized to related reduction mechanisms, including squeezing [Elvira and Herzet, 2020], relaxing [Guyard et al., 2022], and peeling [Guyard et al., 2023].

[LE: Will Be Verified.] This paper is concerned with a different axis of classification, namely the screening target: *feature screening* and *sample screening*. Rather than studying the safety conditions guaranteeing that a screening rule preserves the optimal solution, we study the mathematical structure underlying screening itself.

[LE: Will Be Verified.] Feature screening was first introduced for sparse learning models such as the Lasso, where inactive features can be identified and removed before optimization [El Ghaoui et al., 2010b]. Sample screening was later developed for the soft-margin support vector machine (SVM) [Ogawa et al., 2014]. Historically, it was not introduced as an independent mathematical operation. Instead, the feature-screening philosophy was transferred to the SVM’s dual problem, where the dual variables correspond to training samples, and the resulting elimination rule was interpreted as sample screening. Since then, feature screening and sample screening have evolved largely in parallel. They are now combined in simultaneous screening methods [Shibagaki et al., 2016], and several recent frameworks derive both through Fenchel–Rockafellar (FR) duality [Ndiaye et al., 2017; Yamada and Yamada, 2021; Tran et al., 2025].

[LE: Will Be Verified.] This historical development suggests a close relationship between feature screening and sample screening. Yet this relationship has never been formulated mathematically. Existing work derives screening rules for particular optimization models rather than defining screening itself as a model-independent mathematical transformation. Consequently, although feature screening and sample screening are widely regarded as primal–dual counterparts, the duality between them remains an informal intuition rather than a precise mathematical statement.

This naturally raises two fundamental questions. Can feature screening and sample screening be formulated as mathematical transformations independent of specific optimization models? If so, in what mathematical sense are they dual?

[LE: Will Be Verified.] We answer these questions by formulating screening as model-independent transformations on a class of convex optimization called *FR representations*. Once such a class is identified, both FR duality and screening can be considered as transformations on the same mathematical objects, making their relationship meaningful and directly comparable. Specifically, various screening methods act on the following convex optimization problems

$$\min_{x \in \mathbb{R}^n} p(x) = f(Ax) + g(x) + a, \quad A \in \mathbb{R}^{m \times n}, \quad a \in \mathbb{R},$$

with separable functions f and g . The scalar component a is included to ensure that this class of representations is stable under the transformations considered in this paper. In machine learning, the columns of A represent features, whereas its rows represent samples. This class encompasses many standard learning models, including the Lasso, Elastic Net [Guyard et al., 2022], logistic regression [Wang et al., 2014], Huber regression [Chen et al., 2020], Kullback–Leibler regression [Dantas et al., 2021], and SVMs [Zhai et al., 2020; Nguyen et al., 2026].

[LE: Will Be Verified.] We encode above objective function as a quadruple $p = (f, g, A, a)$, called a FR representation, and denote by Γ the class of all FR representations. It is important to note that FR duality, say $(\cdot)^*$, is also a transformation on Γ .

Feature screening, restated and generalized in this paper, assumes that a block of variables satisfies $x_{\bar{I}} = z$, where $I \sqcup \bar{I} = \{1, \dots, n\}$, and transforms the function p into¹

$$S^F(p) = f(A_I x_I + A_{\bar{I}} z) + g_I(x_I) + a + g_{\bar{I}}(z).$$

The classical feature screening is recovered by the special case $z = 0$, see e.g., [El Ghaoui et al., 2010a].

Sample screening, restated and generalized in this paper, assumes $s \in \partial f_{\bar{J}}(A_{\bar{J}} x)$, where $J \sqcup \bar{J} = \{1, \dots, m\}$. By the Fenchel–Young equality, the loss function f at block $\bar{J}$ can be replaced by an affine function, $f_{\bar{J}}(A_{\bar{J}} x) = \langle s, A_{\bar{J}} x \rangle - f_{\bar{J}}^*(s)$, the function p becomes

$$S^S(p) = f_J(A_J x) + g(x) + \langle A_{\bar{J}}^\top s, x \rangle + a - f_{\bar{J}}^*(s).$$

The classical sample screening for soft-margin SVM [Ogawa et al., 2013] is recovered by the special case $s = 0$.

[LE: Will Be Verified.] We now regard S^F , S^S , and $(\cdot)^*$ as transformations on Γ . We show that both screening transformations admit the same structural decomposition into a translation followed by a restriction, while the primitive transformations are *equivariant* under FR duality. These two structural results provide the key ingredients for establishing the equivariance of screening:

$$(S^F(p))^* = S^S(p^*).$$

This identity is the mathematical formulation of the duality between feature screening and sample screening: feature screening on a primal problem and sample screening on its dual problem provide two problems related via FR duality. The duality between two screening transformations can be expressed by the commutative diagram

$$\begin{array}{ccc} p & \xrightarrow{S^F} & S^F(p) \\ * \downarrow & & \downarrow * \\ p^* & \xrightarrow{S^S} & S^S(p^*) \end{array}$$

[LE: Will Be Verified.] This paper makes two main contributions. First, we formulate feature screening and sample screening as model-independent transformations on FR representations, providing a common mathematical language for screening beyond individual optimization models. Second, we provide a precise mathematical formulation of the duality between feature screening and sample screening via equivariance and FR duality. Within this framework, the intrinsic complexity reduction and simultaneous screening follow as immediate consequences. Our proposed framework also provides a foundation for future extensions to safe screening methods.

¹Throughout the introduction we omit the subscript parameters of S^F and S^S for readability.

[LE: Will Be Verified.] The rest of the paper is organized as follows. Section 2 introduces FR representations together with the primitive transformations including FR duality, translation and restriction acting on them, providing the common mathematical environment for screening. Section 3 then defines feature screening and sample screening as transformations on this class of representations, proves that both decompose into translation and restriction, and establishes the duality between feature and sample screening.

2 FR Representations and Equivariant Transformations

This section does not study screening directly. Instead, it introduces FR representations together with the primitive transformations acting on them: FR duality, translation, and restriction. We also establish that translation and restriction are equivariant under FR duality.

Throughout this section, functions take values in the extended reals $\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\}$, so that a domain constraint can be encoded as $+\infty$ outside the feasible set. For a finite set N and $I \subseteq N$, $\bar{I} := N \setminus I$ denotes its complement, and disjoint-union $I \sqcup \bar{I} = N$ records that I and $\bar{I}$ partition N . This disjoint-union notation is used for both index sets fixed below: $I \sqcup \bar{I} = \{1, \dots, n\}$ and $J \sqcup \bar{J} = \{1, \dots, m\}$.

2.1 FR Representations

We consider the problems of the form

$$\min_{x \in \mathbb{R}^n} p(x), \quad p(x) = f(Ax) + g(x) + a. \quad (1)$$

where

$$f : \mathbb{R}^m \rightarrow \overline{\mathbb{R}}, \quad g : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}, \quad A \in \mathbb{R}^{m \times n}, \quad a \in \mathbb{R}, \quad (2)$$

with f and g closed, proper, and convex. We further assume that f and g are separable, $f(y) = \sum_{j=1}^m f_j(y_j)$ and $g(x) = \sum_{i=1}^n g_i(x_i)$, with each f_j and g_i closed, proper, and convex on $\mathbb{R}$. Here, at the first glance, a is a redundant constant since it does not affect the minimizer of p . However, it is important to keep track of a because it is used as an absorbing constant in the following transformations.

[LE: Will Be add ref]. The problem (1) is of particular importance in both convex optimization and machine learning. In convex optimization, it forms the foundation of FR duality theory. In machine learning, it provides a unified formulation for a wide range of learning problems, including regression, classification, and optimal transport.

Definition 2.1 (FR representation). If (f, g, A, a) satisfies (2), then we say that it is well-defined. In this case, we refer to it as an FR representation of function p .

In this paper, we identify the optimization problem (1) with its objective function p and its FR representation (f, g, A, a) . An FR representation is not merely a convenient way to write down a function. The quadruple (f, g, A, a) is designed so that the class of FR representations is *closed* under the three transformations studied in this section.

For function p in (1), we write $\text{size}(p) := (m, n)$ the *problem size*. We also refer to it as the *intrinsic complexity* of the problem. We define $\Gamma_{m,n}$ to be the set of all functions p with problem size (m, n) , and Γ to be the set of all FR representations of any size. If both f and g are separable, then (f, g, A, a) is said to be a *separable FR representation*. Intrinsic complexity and separability are essential and will be discussed in more details in the next section.

2.2 FR Duality

For a function $\varphi : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$, its Fenchel conjugate is

$$\varphi^*(y) := \sup_{x \in \mathbb{R}^d} \{\langle y, x \rangle - \varphi(x)\}.$$

When φ is closed, proper, and convex, then $\varphi^{**} = \varphi$, a fact used repeatedly below. [LE: Will Be add ref.]

The FR dual problem of (1) is

$$\max_{y \in \mathbb{R}^m} -d(y), \quad d(y) = g^*(-A^\top y) + f^*(y) - a.$$

Note that in the classical FR duality, we do not have a and $-a$ in p and d . Here the constant a is negated in the dual, so that the weak duality inequality $p(x) + d(y) \geq 0$ remains valid for all $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$.

The dual function d is itself an FR representation, with the problem size (n, m) . Therefore, FR duality can also be considered as a transformation on the class of FR representations. This motivates the following definition.

Definition 2.2 (FR Duality). The FR duality is a transformation $(\cdot)^* : \Gamma_{m,n} \rightarrow \Gamma_{n,m}$ such that the image of $p = (f, g, A, a)$ is

$$p^* := (g^*, f^*, -A^\top, -a).$$

Note that FR duality is also a transformation on the class of FR representations Γ .

[LE: Will Be Verified.]

Proposition 2.3 (FR Duality Involution). *For every FR representation $p \in \Gamma$, we have $(p^*)^* = p$.*

Proof. Write $p = (f, g, A, a)$, so $p^* = (g^*, f^*, -A^\top, -a)$. Applying the same rule again,

$$(p^*)^* = ((f^*)^*, (g^*)^*, -(-A^\top)^\top, -(-a)) = (f^{**}, g^{**}, A, a) = (f, g, A, a) = p,$$

using biconjugation $f^{**} = f$, $g^{**} = g$, which hold true since f and g are closed, proper, and convex. $\square$

FR duality is itself a transformation on Γ . It naturally induces a notion of duality between transformations, called *equivariance*, this is the main structure to investigate in the remaining paper.

Definition 2.4 (Equivariance under FR duality). Transformations $F, G : \Gamma \rightarrow \Gamma$ are *equivariant under FR duality* if

$$(F(p))^* = G(p^*), \quad \forall p \in \Gamma.$$

2.3 Translation

We first introduce three primitive translations.

Definition 2.5 (Primitive translations). For $b \in \mathbb{R}^d$, $c \in \mathbb{R}$, and $d \in \mathbb{R}^d$, define, for a function $\varphi : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$,

$$\text{domain translation: } (T_b^d \varphi)(y) := \varphi(y + b),$$

$$\text{value translation: } (T_c^v \varphi)(y) := \varphi(y) + c,$$

$$\text{slope translation: } (T_d^s \varphi)(x) := \varphi(x) + \langle d, x \rangle.$$

Geometrically, domain translation shifts the graph horizontally, value translation shifts it vertically, and slope translation tilts the graph by adding a linear function, thereby shifting every subgradient by the fixed vector d .

Definition 2.6 (Translation parameter and its space). We call $K_{m,n} := \mathbb{R}^m \times \mathbb{R} \times \mathbb{R}^n$, a *translation parameter space*. An element $k = (b, c, d) \in K_{m,n}$ is called a *translation parameter*.

Now, we can define a translation indexed by a translation parameter.

Definition 2.7 (Translation). Let $p = (f, g, A, a) \in \Gamma_{m,n}$. The translation T_k indexed by $k = (b, c, d) \in K_{m,n}$ on p is defined by $T_k : \Gamma_{m,n} \rightarrow \Gamma_{m,n}$ such that

$$T_k(p) := (T_b^d f, T_d^s g, A, a + c).$$

Note that, the matrix A is untouched by translation: only the two functions and the constant move. This is what lets restriction and translation later be composed without the matrix itself being disturbed by which primitive acts first.

It is clear that the family $\{T_k\}$ forms an Abelian translation action on the class of FR representations:

$$T_{k_1+k_2}(p) = T_{k_1}(T_{k_2}(p)) = T_{k_2}(T_{k_1}(p)).$$

[LE: Will Be verified. But kept because of its beauty. So keep it of this form, do not replace it by a proposition. The commutative will be used in the next paper.]

Section 3 will apply the translation action after choosing k as a function of the coordinates being screened. Before that, we establish this subsection's law: how the translation action interacts with FR duality.

Proposition 2.8 (Primitive Translation Equivariance). *For every convex function $\varphi : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$, $b, d \in \mathbb{R}^n$, and $c \in \mathbb{R}$, we have*

1. $(T_b^d \varphi)^* = T_{-b}^s(\varphi^*)$
2. $(T_d^s \varphi)^* = T_{-d}^d(\varphi^*)$
3. $(T_c^v \varphi)^* = T_{-c}^v(\varphi^*)$

Proof. [LE: Will Be TODO. Add ref.] □

[LE: Will Be Verified.]

We define the *dual parameter* of k by $k^* = (-d, -c, -b) \in K_{n,m}$. Intuitively, the duality map $k \mapsto k^*$ swaps the domain and slope translation parameters, and negates all three parameters. Thus, $*$: $K_{m,n} \rightarrow K_{n,m}$ is a duality map such that $(k^*)^* = k$.

Proposition 2.9 (Translation Equivariance). *Let $p \in \Gamma_{m,n}$ be an FR representation and let $k = (b, c, d) \in K_{m,n}$ be a translation parameter. Then*

$$(T_k(p))^* = T_{k^*}(p^*).$$

Proof. Write $p = (f, g, A, a)$, so $T_k(p) = (T_b^d f, T_d^s g, A, a + c)$. Applying the FR dual, which negates the constant slot, $(f_0, g_0, A_0, a_0)^* = (g_0^*, f_0^*, -A_0^\top, -a_0)$, termwise, and using Proposition 2.8,

$$(T_k(p))^* = ((T_d^s g)^*, (T_b^d f)^*, -A^\top, -(a + c)) = (T_{-d}^d(g^*), T_{-b}^s(f^*), -A^\top, -a - c).$$

On the other side, $p^* = (g^*, f^*, -A^\top, -a)$, and $k^* = (-d, -c, -b)$, so $T_{k^*}(p^*)$ applies a domain translation by $-d$ to p^* 's first slot, a slope translation by $-b$ to its second slot, and adds $-c$ to its constant:

$$T_{k^*}(p^*) = (T_{-d}^d(g^*), T_{-b}^s(f^*), -A^\top, -a - c).$$

Every term matches $(T_k(p))^*$, which proves the theorem. $\square$

[LE: Will Be Verified.]

2.4 Restriction

Let $I \subseteq \{1, \dots, n\}$ and $J \subseteq \{1, \dots, m\}$. For a separable function $g(x) = \sum_{i=1}^n g_i(x_i)$, write $x_I := (x_i)_{i \in I}$ and $g_I(x_I) := \sum_{i \in I} g_i(x_i)$; define f_J symmetrically for a separable $f(y) = \sum_{j=1}^m f_j(y_j)$. For a matrix $A \in \mathbb{R}^{m \times n}$, write $A_I := A_{:,I}$ for the submatrix of columns indexed by I , and $A_J := A_{J,:}$ for the submatrix of rows indexed by J . Combining both, $A_{J,I} := (A_J)_I = (A_I)_J$ denotes the submatrix with rows in J and columns in I .

Definition 2.10 (Feature and sample restriction). Let $I \sqcup \bar{I} = \{1, \dots, n\}$ and $J \sqcup \bar{J} = \{1, \dots, m\}$. Define the feature restriction operator $R_I^F : \Gamma_{m,n} \rightarrow \Gamma_{m,|I|}$ and the sample restriction operator $R_J^S : \Gamma_{m,n} \rightarrow \Gamma_{|J|,n}$ by

$$R_I^F(f, g, A, a) := (f, g_I, A_I, a).$$

and

$$R_J^S(f, g, A, a) := (f_J, g, A_J, a).$$

Feature restriction only touches g and the columns of A ; sample restriction only touches f and the rows of A . Because they act on disjoint parts of p , the order in which they are applied cannot matter: this subsection's law.

Proposition 2.11 (Restriction Equivariance). *For every $p = (f, g, A, a) \in \Gamma$, every $I \subseteq \{1, \dots, n\}$, and every $J \subseteq \{1, \dots, m\}$,*

$$(R_I^F(p))^* = R_I^S(p^*), \quad (R_J^S(p))^* = R_J^F(p^*).$$

Proof. Write $p^* = (g^*, f^*, -A^\top, -a)$. Applying the FR dual termwise to $R_I^F(p) = (f, g_I, A_I, a)$,

$$(R_I^F(p))^* = ((g_I)^*, f^*, -A_I^\top, -a).$$

By separability of g , $(g_I)^* = (g^*)_I$. Rows I of $-A^\top$ equal $-A_I^\top$, since rows of $A^\top$ are columns of A transposed. Hence

$$(R_I^F(p))^* = ((g^*)_I, f^*, -A_I^\top, -a) = R_I^S(p^*),$$

the last equality by the definition of sample restriction applied to $p^* = (g^*, f^*, -A^\top, -a)$. The second identity is the mirror computation, restricting $R_J^S(p) = (f_J, g, A_J, a)$ instead, and using $(f_J)^* = (f^*)_J$ by the same separability argument. $\square$

[LE: Will Be Verified.]

3 Duality of Feature and Sample Screening

This section shows that feature screening and sample screening form an equivariant pair under FR duality. We proceed in four steps. First, we give a formal definition of feature and sample screening, independent of any specific application problem. Second, we define feature and sample translation and establish their equivariance. Third, we show that both screenings decompose into a translation followed by a restriction. Finally, combining these two facts, we prove the main duality identity. As a by-product, we also discuss the resulting reduction in intrinsic complexity and a characterization of simultaneous screening.

3.1 Feature and Sample Screening as Transformations

This subsection defines feature screening and sample screening as transformations on Γ . Recall that an FR representation $p = (f, g, A, a)$ encodes the objective function

$$p(x) = f(Ax) + g(x) + a.$$

In the literature, feature screening typically eliminates a block of primal variables. We state the assumption slightly more generally, allowing the eliminated block to sit at an arbitrary fixed point rather than requiring it to be zero. Let $I \sqcup \bar{I} = \{1, \dots, n\}$ and $z \in \mathbb{R}^{\bar{I}}$ be a known vector, and suppose that

$$x_{\bar{I}} = z.$$

Writing $x = (x_I, x_{\bar{I}})$, $A = (A_I, A_{\bar{I}})$, and $g(x) = g_I(x_I) + g_{\bar{I}}(x_{\bar{I}})$ by separability, substituting $x_{\bar{I}} = z$ into the objective gives

$$p(x_I, z) = f(A_I x_I + A_{\bar{I}} z) + g_I(x_I) + a + g_{\bar{I}}(z).$$

Definition 3.1 (Feature screening). The *feature screening* of p at (I, z) is the FR representation

$$S_{I,z}^F(p) := (T_{A_{\bar{I}}z}^d f, g_I, A_I, a + g_{\bar{I}}(z)).$$

Here, $S_{I,z}^F(p)(x_I) = p(x_I, z)$. This means the screened objective is exactly the original objective with $x_{\bar{I}}$ fixed at z .

Example 3.2 (Feature screening for the Lasso). Classical feature screening for the Lasso is the special case $z = 0$ of the proposed feature screening operator $S_{I,z}^F$. The Lasso problem

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1, \quad \lambda > 0,$$

is an FR representation $p = (f, g, A, 0)$ with $f(y) = \frac{1}{2} \|y - b\|_2^2$ and $g(x) = \lambda \|x\|_1$.

In sparse learning, feature screening certifies that a block of coefficients is inactive at the optimum. For the Lasso, inactivity means that the corresponding coefficients vanish,

$$x_{\bar{I}} = 0,$$

exactly the special case $z = 0$ of the general feature-screening assumption $x_{\bar{I}} = z$. Since $z = 0$, the screened function $S_{I,0}^F(p)$ is simply the restriction of the Lasso representation to the remaining coordinates, with corresponding problem

$$\min_{x_I \in \mathbb{R}^{|I|}} \frac{1}{2} \|A_I x_I - b\|_2^2 + \lambda \|x_I\|_1.$$

The proposed definition thus recovers classical Lasso feature screening, while extending it from coefficients certified to be zero to coefficients certified to take any fixed value z .

Feature screening simplifies the objective function by fixing a block of the variables in x . Sample screening simplifies the objective function by fixing a block of the loss function f . Let $J \sqcup \bar{J} = \{1, \dots, m\}$ and $s \in \mathbb{R}^{\bar{J}}$, and suppose that

$$s \in \partial f_{\bar{J}}(A_{\bar{J}}x),$$

i.e., s is a fixed subgradient of the block $\bar{J}$ of f evaluated at x . By the Fenchel–Young equality, which holds in $f(y) + f^*(s) \geq \langle s, y \rangle$ exactly when $s \in \partial f(y)$, the corresponding block of f can be replaced by a simple affine function, i.e.,

$$f_{\bar{J}}(A_{\bar{J}}x) = \langle s, A_{\bar{J}}x \rangle - f_{\bar{J}}^*(s).$$

Substituting this into the objective gives

$$p(x) = f_J(A_Jx) + g(x) + \langle A_J^\top s, x \rangle + a - f_{\bar{J}}^*(s).$$

Definition 3.3 (Sample screening). The *sample screening* of p at (J, s) is the FR representation

$$S_{J,s}^S(p) := (f_J, T_{A_J^\top s}^s g, A_J, a - f_{\bar{J}}^*(s)),$$

so that $S_{J,s}^S(p)(x)$ equals the original objective with $f_{\bar{J}}(A_{\bar{J}}x)$ replaced by its Fenchel–Young-tight value $\langle s, A_{\bar{J}}x \rangle - f_{\bar{J}}^*(s)$.

Geometrically, sample screening replaces the eliminated loss block by its exact affine value at the certified dual point s .

Example 3.4 (Sample screening for the soft-margin SVM). Classical sample screening for the soft-margin SVM is the special case $s = 0$ of the proposed sample screening operator $S_{J,s}^S$. The soft-margin SVM problem

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|x\|_2^2 + C \sum_{i=1}^m \max(0, 1 - b_i \langle a_i, x \rangle), \quad C > 0,$$

is represented in FR form by $p = (f, g, A, 0)$, where $f(y) = C \sum_{i=1}^m \max(0, 1 - b_i y_i)$, $g(x) = \frac{1}{2} \|x\|_2^2$, and $A = (a_1, \dots, a_m)^\top$.

From [Ogawa et al., 2013], the idea of sample screening is that, if one can identify a sample i that is correctly classified, i.e.,

$$1 - b_i \langle a_i, x \rangle < 0,$$

then one can remove the corresponding loss term from the loss function.

For $J \sqcup \bar{J} = \{1, \dots, m\}$, applying the above sample screening condition for all indices $i \in \bar{J}$ can be rewritten as the following condition²

$$0 \in \partial f_{\bar{J}}(A_{\bar{J}}x).$$

This is exactly the special case $s = 0$ of the generalized assumption $s \in \partial f_{\bar{J}}(A_{\bar{J}}x)$. In this case, the translation terms vanish, $A_J^\top s = 0$ and $f_{\bar{J}}^*(s) = 0$, so $S_{J,0}^S(p) = (f_J, g, A_J, 0)$, and the screened problem is

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|x\|_2^2 + C \sum_{i \in J} \max(0, 1 - b_i \langle a_i, x \rangle).$$

²In the original paper [Ogawa et al., 2013], the authors also screen samples such that $1 \in \partial f_{\bar{J}}(A_{\bar{J}}x)$. Our sample screening operator also covers this case; for simplicity, we do not discuss it in this example.

The proposed definition thus recovers classical SVM sample screening, while extending it from samples certified to satisfy $0 \in \partial f_i$ to samples certified to satisfy an arbitrary subgradient condition $s \in \partial f_i$.

Both definitions already display the matrix slot explicitly, so the effect of screening on problem size needs no further argument.

Corollary 3.5 (Screening reduces problem size). *For every $p = (f, g, A, a) \in \Gamma_{m,n}$, every $I \sqcup \bar{I} = \{1, \dots, n\}$, and every $J \sqcup \bar{J} = \{1, \dots, m\}$,*

$$\text{size}(S_{I,z}^F(p)) = (m, |I|), \quad \text{size}(S_{J,s}^S(p)) = (|J|, n).$$

Proof. Immediate from the definitions above: $S_{I,z}^F(p)$ has matrix slot $A_I \in \mathbb{R}^{m \times |I|}$, and $S_{J,s}^S(p)$ has matrix slot $A_J \in \mathbb{R}^{|J| \times n}$. $\square$

Reading intrinsic complexity as problem size, screening rules reduce the intrinsic complexity. Theorem 3.9 below explains why this must always happen, since translation never changes the shape of A and restriction is the only step that can, but the size claim itself needs nothing beyond the definitions just given.

Both definitions arose from a direct assumption on p , a primal value for feature screening and a dual subgradient for sample screening, with no reference to translation or restriction. Yet both formulas visibly split into "shift f or g , then drop coordinates." Section 2 introduced exactly two primitives for this: translation (T^d , T^v , T^s , combined into T_k) and restriction (R^F , R^S). Theorem 3.9 shows that $S_{I,z}^F$ and $S_{J,s}^S$, defined above purely from optimization assumptions, are always built from exactly these two primitives and nothing else.

3.2 Feature–Sample Translation

In Subsection 2.3, we introduced the translation T_k and its translation parameter k . In this subsection, we introduce feature and sample translation, built on top of T_k and k . Surprisingly, these objects also admit equivariance under FR duality. These notions play an important role in establishing the main results proved in the following subsections.

Definition 3.6 (Feature–Sample Translation). We define the feature translation

$$T_{I,z}^F(p) = T_k(p), \quad k = \kappa_{I,z}^F(p) := (A_{\bar{I}}z, g_{\bar{I}}(z), 0) \in K_{m,|\bar{I}|}.$$

where k is a feature translation parameter depending on p , index set I and $z \in \mathbb{R}^{\bar{I}}$.

Symmetrically, we define the sample translation

$$T_{J,s}^S(p) = T_h(p), \quad h = \kappa_{J,s}^S(p) := (0, -f_{\bar{J}}^*(s), A_{\bar{J}}^\top s) \in K_{|\bar{J}|,n}.$$

where h is a sample translation parameter depending on p , index set J and $s \in \mathbb{R}^{\bar{J}}$.

Now we show that there is an equivariance structure on both the parameter level and the translation level.

Proposition 3.7 (Parameter Equivariance). *For every $p = (f, g, A, a) \in \Gamma_{m,n}$, every $I \sqcup \bar{I} = \{1, \dots, n\}$, $z \in \mathbb{R}^{\bar{I}}$, every $J \sqcup \bar{J} = \{1, \dots, m\}$, and $s \in \mathbb{R}^{\bar{J}}$,*

$$\begin{aligned} (\kappa_{I,z}^F(p))^* &= \kappa_{I,z}^S(p^*), \\ (\kappa_{J,s}^S(p))^* &= \kappa_{J,s}^F(p^*). \end{aligned}$$

Proof. Using $k^* = (-d, -c, -b)$ for $k = (b, c, d)$ (Section 2),

$$(\kappa_{I,z}^F(p))^* = (A_{\bar{I}}z, g_{\bar{I}}(z), 0)^* = (0, -g_{\bar{I}}(z), -A_{\bar{I}}z).$$

Write $p^* = (g^*, f^*, -A^\top, -a)$. By definition, the sample translation parameter of p^* at (I, z) is

$$\kappa_{I,z}^S(p^*) = (0, -(g^*)_{\bar{I}}^*(z), (-A^\top)_{\bar{I}}^\top z).$$

Since g is separable, its conjugate restricts termwise, $(g^*)_{\bar{I}} = (g_{\bar{I}})^*$, so $(g^*)_{\bar{I}}^*(z) = (g_{\bar{I}})^{**}(z) = g_{\bar{I}}(z)$ by biconjugation, using that $g_{\bar{I}}$, a finite sum of closed, proper, convex functions, is itself closed, proper, convex[LE: Will Be Basic convex-analysis fact, citation to follow.]. Rows $\bar{I}$ of $-A^\top$ equal $-(A_{\bar{I}})^\top$, so $(-A^\top)_{\bar{I}}^\top z = -A_{\bar{I}}z$. Hence

$$\kappa_{I,z}^S(p^*) = (0, -g_{\bar{I}}(z), -A_{\bar{I}}z) = (\kappa_{I,z}^F(p))^*.$$

We now prove the second identity, which does not follow from the first by simply exchanging f and g : Definition 3.6 already builds $\kappa_{I,z}^F$ and $\kappa_{J,s}^S$ asymmetrically, so it needs its own computation. Using $k^* = (-d, -c, -b)$ again,

$$(\kappa_{J,s}^S(p))^* = (0, -f_{\bar{J}}^*(s), A_{\bar{J}}^\top s)^* = (-A_{\bar{J}}^\top s, f_{\bar{J}}^*(s), 0).$$

Write $p^* = (g^*, f^*, -A^\top, -a) \in \Gamma_{n,m}$. Since $J \subseteq \{1, \dots, m\}$ indexes the second slot's problem size for p^* , the feature translation parameter of p^* at (J, s) is

$$\kappa_{J,s}^F(p^*) = ((-A^\top)_{J^\top} s, (f^*)_{J^\top}(s), 0),$$

by the same definition as $\kappa_{I,z}^F$, with g^* taking the role of the first slot's function and $-A^\top$ the role of the matrix. Columns $\bar{J}$ of $-A^\top$ equal $-(A_{\bar{J}})^\top$, since columns of $A^\top$ are rows of A transposed, so $(-A^\top)_{\bar{J}} s = -A_{\bar{J}}^\top s$. Since f is separable, its conjugate restricts termwise, $(f^*)_{\bar{J}} = (f_{\bar{J}})^*$, so $(f^*)_{\bar{J}}(s) = f_{\bar{J}}^*(s)$; this is the same separability fact used above, now applied to f rather than g , and used directly rather than through biconjugation, since here it is the conjugate f^* itself, not its biconjugate, that must be restricted. Hence

$$\kappa_{J,s}^F(p^*) = (-A_{\bar{J}}^\top s, f_{\bar{J}}^*(s), 0) = (\kappa_{J,s}^S(p))^*.$$

Unlike the first identity, this computation never invokes biconjugation: $\kappa_{J,s}^S(p)$ already carries a conjugate, $f_{\bar{J}}^*(s)$, in its second slot, so dualizing and restricting commute with a single separability step, not two. $\square$

Proposition 3.7 identifies where the feature-sample asymmetry lives. Feature screening produces a domain shift and a value shift, while sample screening produces a slope shift and a conjugate value shift. The proposition shows that this apparent asymmetry is fully absorbed by the canonical translation parameters: FR duality exchanges the feature parameter with the sample parameter. The next step is to lift this parameter equivariance from parameters to translation operators.

Proposition 3.8 (Feature-Sample Translation Equivariance). *For every $p = (f, g, A, a) \in \Gamma_{m,n}$, every $I \sqcup \bar{I} = \{1, \dots, n\}$, $z \in \mathbb{R}^{\bar{I}}$, every $J \sqcup \bar{J} = \{1, \dots, m\}$, and $s \in \mathbb{R}^{\bar{J}}$,*

$$\begin{aligned} (T_{I,z}^F(p))^* &= T_{I,z}^S(p^*), \\ (T_{J,s}^S(p))^* &= T_{J,s}^F(p^*). \end{aligned}$$

Proof. By Translation Equivariance (Proposition 2.9, Section 2) applied to $k = \kappa_{I,z}^F(p)$,

$$(T_{I,z}^F(p))^* = (T_{\kappa_{I,z}^F(p)}(p))^* = T_{(\kappa_{I,z}^F(p))^*}(p^*).$$

By Parameter Equivariance (Proposition 3.7), $(\kappa_{I,z}^F(p))^* = \kappa_{I,z}^S(p^*)$, so

$$(T_{I,z}^F(p))^* = T_{\kappa_{I,z}^S(p^*)}(p^*) = T_{I,z}^S(p^*).$$

The second identity follows the same argument, exchanging the roles of f and g , I and J , and domain and slope translation. $\square$

3.3 Decomposition of Screening

This subsection proves that feature screening and sample screening are not primitive: each decomposes into a translation followed by a restriction.

Theorem 3.9 (Screening Decomposition). *For every $p = (f, g, A, a)$, every $I \sqcup \bar{I} = \{1, \dots, n\}$, $z \in \mathbb{R}^{\bar{I}}$, $J \sqcup \bar{J} = \{1, \dots, m\}$, and $s \in \mathbb{R}^{\bar{J}}$,*

$$S_{I,z}^F = R_I^F \circ T_{I,z}^F, \quad S_{J,s}^S = R_J^S \circ T_{J,s}^S.$$

Proof. By Definition 3.6, $\kappa_{I,z}^F(p)$ has zero slope component, so

$$T_{I,z}^F(p) = (T_{A_{\bar{I}}z}^d f, g, A, a + g_{\bar{I}}(z)).$$

Applying R_I^F , which restricts the second and third slots to I and leaves the first and fourth untouched,

$$R_I^F(T_{I,z}^F(p)) = (T_{A_{\bar{I}}z}^d f, g_I, A_I, a + g_{\bar{I}}(z)),$$

which is exactly $S_{I,z}^F(p)$ from the definition above.

The sample case is the mirror computation. Since $\kappa_{J,s}^S(p)$ has zero domain component,

$$T_{J,s}^S(p) = (f, T_{A_{\bar{J}}s}^s g, A, a - f_{\bar{J}}^*(s)).$$

Applying R_J^S , which restricts the first and third slots to J and leaves the second and fourth untouched,

$$R_J^S(T_{J,s}^S(p)) = (f_J, T_{A_{\bar{J}}s}^s g, A_J, a - f_{\bar{J}}^*(s)),$$

which is exactly $S_{J,s}^S(p)$. $\square$

[LE: Will Be Verified.]

This is the paper's central structural fact, screening equals translation followed by restriction, now proved rather than assumed. Feature screening and sample screening were defined from two unrelated-looking optimization assumptions, a primal fixed point and a dual subgradient, and Theorem 3.9 shows they are the same two-step construction, one primitive translation and one primitive restriction, applied to the two different primitives of Section 2. This is what lets the rest of this section's results, Feature-Sample Duality and Simultaneous Screening Equivariance, follow from Section 2's Translation Equivariance and Restriction Equivariance, rather than needing a separate argument for each screening type.

Theorem 3.9 also gives a second proof of Corollary 3.5, stated already above directly from the definitions: since translation never changes the shape of A (Section 2), the decomposition shows the size change comes entirely from the restriction half, R_I^F or R_J^S , consistent with, and not needed beyond, the direct argument already given.

3.4 Duality of Screening

This subsection proves that feature screening and sample screening are equivariant under FR duality: dualizing one produces exactly the other, on the dual representation. Feature screening moves the eliminated block of x into a domain translation of f , then discards it from g . FR duality exchanges domain translations with slope translations and exchanges the roles of f and g (Section 2). Sample screening moves an eliminated block of y into a slope translation of g , then discards it from f . Matching these two descriptions suggests that dualizing a feature screening produces exactly a sample screening, with no other operation involved.

Theorem 3.10 (Feature–Sample Duality). *Let $p = (f, g, A, a) \in \Gamma_{m,n}$. For every $I \sqcup \bar{I} = \{1, \dots, n\}$ and $z \in \mathbb{R}^{\bar{I}}$,*

$$(S_{I,z}^F(p))^* = S_{I,z}^S(p^*).$$

Symmetrically, for every $J \sqcup \bar{J} = \{1, \dots, m\}$ and $s \in \mathbb{R}^{\bar{J}}$,

$$(S_{J,s}^S(p))^* = S_{J,s}^F(p^*).$$

Proof. We prove the first identity; the second is the mirror computation with the roles of f and g , I and J , domain and slope translation, exchanged. The proof combines Screening Decomposition (Theorem 3.9, this section) with Restriction Equivariance (Section 2) and Feature–Sample Translation Equivariance (Proposition 3.8, this section).

By Theorem 3.9, $S_{I,z}^F(p) = R_I^F(T_{I,z}^F(p))$. By Restriction Equivariance, applied to $q := T_{I,z}^F(p)$,

$$(S_{I,z}^F(p))^* = (R_I^F(q))^* = R_I^S(q^*) = R_I^S((T_{I,z}^F(p))^*).$$

By Feature–Sample Translation Equivariance, $(T_{I,z}^F(p))^* = T_{I,z}^S(p^*)$. Substituting,

$$(S_{I,z}^F(p))^* = R_I^S(T_{I,z}^S(p^*)) = S_{I,z}^S(p^*),$$

the last equality by Theorem 3.9 applied to p^* . This proves the first identity.

The second identity is the mirror computation, exchanging the roles of f and g , I and J , and domain and slope translation, using the second halves of Screening Decomposition, Restriction Equivariance, and Feature–Sample Translation Equivariance. $\square$

Theorem 3.10 says a single screening dualizes to a single screening of the other type. What happens when a feature screening and a sample screening are both applied to the same p ? This composition was used in [Shibagaki et al., 2016] during the solving process, where it is referred to as simultaneous screening. The following result shows that, by exchanging their order with the appropriate parameters, the two compositions also form an equivariant pair.

Corollary 3.11 (Simultaneous Screening Equivariance). *For every $p = (f, g, A, a) \in \Gamma_{m,n}$, every $I \sqcup \bar{I} = \{1, \dots, n\}$, $z \in \mathbb{R}^{\bar{I}}$, $J \sqcup \bar{J} = \{1, \dots, m\}$, and $s \in \mathbb{R}^{\bar{J}}$, whenever both compositions below are defined,*

$$(S_{J,s}^S S_{I,z}^F(p))^* = S_{J,s}^F S_{I,z}^S(p^*),$$

equivalently,

$$S_{J,s}^S S_{I,z}^F(p) = (S_{J,s}^F S_{I,z}^S(p^*))^*.$$

Proof. Write $q := S_{I,z}^F(p)$. Applying Theorem 3.10 to q and p , respectively, we obtain

$$(S_{J,s}^S(q))^* = S_{J,s}^F(q^*), \quad q^* = (S_{I,z}^F(p))^* = S_{I,z}^S(p^*).$$

Substituting,

$$(S_{J,s}^S S_{I,z}^F(p))^* = S_{J,s}^F(S_{I,z}^S(p^*)) = S_{J,s}^F S_{I,z}^S(p^*),$$

which is the first identity. The second follows by applying $(\cdot)^*$ to both sides and using the involution $(p^*)^* = p$ (Proposition 2.3). $\square$

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